

Study Guide: Chromosomal Basis of Inheritance and Variation

This study guide provides a comprehensive overview of the principles of inheritance and the chromosomal basis of genetics, synthesizing findings from Mendelian experiments to modern chromosomal theory and genetic disorders.

1. Foundations of Genetics: Mendelian Principles

Genetics is the study of heredity—the process by which characters are passed from parent to progeny—and variation, which is the degree to which progeny differ from their parents.

Gregor Johann Mendel's Experiments

Mendel, the "Father of Genetics," conducted hybridization experiments on the garden pea (*Pisum sativum*) for seven years (1856–1863). His success was attributed to studying one character at a time, using true-breeding lines, and applying mathematics/statistics to his results.

Table: Seven Contrasting Characters in Garden Peas

Character	Dominant Trait	Recessive Trait
Stem Height	Tall	Dwarf
Flower Colour	Violet	White
Flower Position	Axial	Terminal
Pod Shape	Inflated	Constricted
Pod Colour	Green	Yellow
Seed Shape	Round	Wrinkled
Seed Colour	Yellow	Green

Mendel's Laws of Inheritance

1. **Law of Dominance:** Characters are controlled by discrete units called factors (genes) which occur in pairs. In a dissimilar pair, one factor dominates the other.
2. **Law of Segregation:** Alleles do not blend; they separate during gamete formation, ensuring each gamete receives only one allele.
3. **Law of Independent Assortment:** When two pairs of traits are combined in a hybrid, the segregation of one pair of characters is independent of the other pair.

2. Chromosomal Theory of Inheritance

Proposed independently by **Walter Sutton and Theodore Boveri (1903)**, this theory united Mendelian principles with cytological observations of chromosome behavior.

Key Historical Developments

- **Wilhelm Roux (1883):** Postulated chromosomes are responsible for heredity.
- **Montgomery (1901):** Suggested distinct pairs of chromosomes exist; maternal chromosomes pair only with paternal ones during meiosis.
- **Sutton and Boveri (1902-1903):** Recognized the parallelism between Mendelian factors and chromosome behavior during gamete formation.

Parallelism Between Genes and Chromosomes

Mendelian Factors (Genes), Chromosome Behaviour

Occur in pairs (Alleles), Occur in pairs (Homologous).

Segregate during gamete formation., Homologous pairs separate during meiosis (Anaphase I).

Assort independently., Separate independently during meiosis.

Experimental Validation: Thomas Hunt Morgan (1910) provided proof for this theory using the fruit fly, *Drosophila melanogaster*. He confirmed that the gene for eye color is located on the X chromosome.

3. Deviations from Mendelian Inheritance

Post-Mendelian discoveries revealed more complex patterns of inheritance:

- **Incomplete Dominance:** Neither allele is completely dominant; the hybrid is an intermediate (e.g., *Mirabilis jalapa* or Snapdragon producing pink flowers from red and white parents).
- **Co-dominance:** Both alleles express themselves equally in the organism (e.g., ABO blood grouping where I^A and I^B alleles are both expressed).
- **Multiple Alleles:** More than two alleles exist for a single gene locus in a population (e.g., the three alleles I^A , I^B , and i in human blood types).

4. Linkage and Crossing Over

Linkage is the physical association of genes on a chromosome. Genes located close together are **linked** and tend to be inherited together, reducing variability.

Types of Linkage

1. **Complete Linkage:** Linked genes are so close that crossing over never occurs; only parental combinations are seen (rare, reported in male *Drosophila*).
2. **Incomplete Linkage:** Linked genes are sufficiently far apart that crossing over is possible, producing both parental and recombinant types (observed in Maize by Hutchinson).

Crossing Over and Recombination

Crossing over is the exchange of segments between non-sister chromatids of homologous chromosomes during the **pachytene** stage of prophase I.

- **Mechanism:** Includes Synapsis (pairing), Tetrad formation, Cross over (at chiasmata), and Terminalization.
- **Recombination:** The process by which DNA segments are broken and recombined to produce new allele combinations. The recombination frequency (RF) is used to construct genetic maps.

5. Sex Determination Mechanisms

Sex determination decides the sexual development of an organism based on various chromosomal strategies:

- **XX-XY Type:** Females are XX (homogametic) and males are XY (heterogametic). Found in humans and *Drosophila*.
- **XX-XO Type:** Females are XX and males are XO (possess only one X). Found in grasshoppers.

- **ZZ-ZW Type:** Males are ZZ (homogametic) and females are ZW (heterogametic). Found in birds.
- **Haplodiploid System:** Found in honey bees. Females (Queen/Workers) are diploid (32 chromosomes) from fertilized eggs. Males (Drones) are haploid (16 chromosomes) developing from unfertilized eggs via **parthenogenesis** (Arrhenotoky).

6. Genetic Disorders

Disorders are classified into two broad categories:

Mendelian Disorders (Single Gene Mutations)

- **Haemophilia:** X-linked recessive "Royal Disease"; results in non-stop bleeding due to a defective clotting protein.
- **Colour Blindness:** X-linked recessive; failure to discriminate between red and green colors.
- **Sickle-Cell Anaemia:** Autosome-linked recessive; substitution of Glutamic acid by Valine at the 6th position of the β -globin chain (GAG to GUG).
- **Phenylketonuria:** Autosomal recessive; lack of enzyme to convert phenylalanine to tyrosine, causing mental retardation.
- **Thalassemia:** Quantitative defect in globin chain synthesis (α on chromosome 16; β on chromosome 11).

Chromosomal Disorders (Numerical/Structural Abnormalities)

- **Aneuploidy:** Gain or loss of individual chromosomes (failure of chromatid segregation).
- **Down's Syndrome:** Trisomy 21 (47 chromosomes). Features include a small round head and furrowed tongue.
- **Klinefelter's Syndrome:** XXY (47 chromosomes). Sterile males with feminine features like Gynaecomastia.
- **Turner's Syndrome:** XO (45 chromosomes). Sterile females with rudimentary ovaries and lack of secondary sexual characters.
- **Polyploidy:** Increase in whole sets of chromosomes (failure of cytokinesis); common in plants.

7. Review Quiz

Short Answer Questions (2-3 sentences each)

1. **Why did Mendel choose the garden pea (*Pisum sativum*) for his experiments?**
2. **Explain the difference between a genotype and a phenotype.**
3. **What is a test cross and why is it used?**
4. **Define "linkage" as coined by T.H. Morgan.**
5. **What are "Fossil Genes"?**
6. **Explain the mechanism of sex determination in honey bees.**
7. **What is the specific molecular cause of Sickle-cell anaemia?**
8. **Differentiate between Aneuploidy and Polyploidy.**

9. What are the primary symptoms of Down's Syndrome?
10. Explain why colour blindness is more common in males than in females.

8. Answer Key

1. Mendel chose the garden pea because it has many distinct contrasting characters and a short life span. Additionally, the flowers normally show self-pollination but are easy to artificially cross-pollinate, and the resulting hybrids are fertile.
2. A genotype refers to the genetic makeup of an organism represented by symbols (e.g., TT, Tt, or tt). A phenotype is the observable physical expression of those genes, such as being "tall" or "dwarf."
3. A test cross is a cross between an individual expressing a dominant trait and a double recessive organism. It is used to determine whether the dominant individual's genotype is homozygous or heterozygous.
4. Linkage describes the physical association or tendency of genes located on the same chromosome to stay together during inheritance. Morgan used this term to explain why certain parental gene combinations appeared more frequently than expected in dihybrid crosses.
5. Fossil genes (or pseudogenes) are sequences in junk DNA that were once functional genes but have lost their ability to make proteins. They serve as "fossilized" records of evolutionary history within the genome.
6. Honey bees use a haplodiploid system where females are diploid (32 chromosomes) and males are haploid (16 chromosomes). Males develop from unfertilized eggs through parthenogenesis, while females develop from fertilized eggs.
7. It is caused by a single base substitution at the sixth codon of the beta-globin gene, changing GAG to GUG. This results in the substitution of the amino acid Glutamic acid with Valine, causing hemoglobin to polymerize under low oxygen.
8. Aneuploidy is the gain or loss of a single chromosome due to the failure of chromatid segregation during cell division. Polyploidy is the increase in a whole set of chromosomes (e.g., 3n, 4n) resulting from the failure of cytokinesis after telophase.
9. Individuals with Down's Syndrome typically have a short stature, a small rounded head, and a broad flat face. Other features include a furrowed big tongue, partially open mouth, and retarded physical and mental development.
10. Colour blindness is an X-linked recessive disorder, meaning the gene is on the X chromosome. Males (XY) only need one defective X to express the trait, whereas females (XX) must inherit two defective X chromosomes to be affected.

9. Essay Format Questions (No Answers Provided)

1. Compare and contrast Mendelian Factors with the behavior of chromosomes, providing the evidence that led to the Chromosomal Theory of Inheritance.
2. Discuss the different types of sex determination systems found in the animal kingdom, citing specific examples for XX/XY, XX/XO, and ZZ/ZW types.
3. Examine the relationship between linkage and crossing over. How do these processes influence genetic variation and the construction of linkage maps?

4. Provide a detailed account of Mendelian disorders, specifically explaining the inheritance patterns and physiological impacts of Haemophilia and Thalassemia.
5. Analyze the role of chromosomal non-disjunction in human health, focusing on the genetic constitution and clinical features of Klinefelter's and Turner's syndromes.

10. Glossary of Key Terms

- **Alleles:** Alternative forms of a gene that occupy the same locus on homologous chromosomes.
- **Bivalent:** A pair of homologous chromosomes aligned side by side during synapsis.
- **Chiasmata:** The points of contact where non-sister chromatids exchange genetic material during crossing over.
- **Emasculation:** The removal of anthers from a flower to prevent self-pollination during hybridization experiments.
- **Gynaecomastia:** The development of breast tissue in males, often observed in Klinefelter's Syndrome.
- **Heterogametic:** An individual producing two different types of gametes regarding sex chromosomes (e.g., human males).
- **Locus:** The specific physical position of a gene on a chromosome.
- **Mutagenic Agents:** Chemical or physical factors that cause mutations in the DNA sequence.
- **Parthenogenesis:** A form of asexual reproduction where an embryo develops from an unfertilized egg.
- **Pedigree Analysis:** The study of a particular trait across several generations of a family tree to determine its inheritance pattern.
- **Recombination Frequency:** The percentage of recombinant offspring in a cross, used to measure the distance between genes.
- **Synapsis:** The intimate pairing of homologous chromosomes during the zygotene stage of meiosis.
- **Synteny:** The condition where two genes are located on the same chromosome, regardless of whether they are linked.
- **Terminalization:** The movement of chiasmata toward the ends of chromatids as homologous chromosomes separate.
- **True Breeding:** A line that undergoes continuous self-pollination and shows stable trait inheritance for several generations.